TEXORA · SESSIONAL NOTES · EXPERIMENT SHEET

Preparation of Stock Solution and Wet Test for Acid Radicals

The sodium-carbonate (soda) extract — why it is prepared, how it is prepared — and the wet confirmatory tests that turn the dry tests' indications into identifications.

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1. Aim and the problem the soda extract solves

Aim: to prepare the stock solution (soda extract) of a given salt and confirm its acid radical by wet tests. Wet tests for anions cannot conveniently be run on the original solution: the cation may precipitate with the test reagent (Ag^+ tests are ruined by many cations), coloured cations mask colours, and insoluble salts will not dissolve at all. The **soda extract** solves all three at once: boiling the salt with sodium carbonate solution converts the salt by double decomposition — the cation leaves as an insoluble carbonate (filtered off), and the anion stays in solution as its soluble sodium salt:

$$CaSO_4 + Na_2CO_3 \rightarrow CaCO_3 \downarrow + Na_2SO_4 \text{ (in solution)} \;;\; \text{in general: salt} + Na_2CO_3 \rightarrow \text{cation carbonate} \downarrow + \text{sodium salt of the anion (filtrate)}$$

The clear filtrate — the soda extract — is then a clean stage on which every anion test can be played. Note the consequence: the extract itself is alkaline and full of carbonate, so portions are **neutralised with the appropriate acid** before tests that carbonate would disturb (and the extract can never be used to test *for* carbonate — that test belongs to the original salt, Experiment 1).

2. Procedure: preparing the soda extract

1. Take about 1 g of the powdered salt in a conical flask or boiling tube with 15–20 mL of distilled water.
2. Add about 3–4 g of pure sodium carbonate (roughly three times the salt) and boil gently for 5–10 minutes, making up the water lost by evaporation.
3. Allow the precipitate to settle; filter while hot; wash the residue with a little hot water, adding the washings to the filtrate.
4. The clear filtrate is the soda extract (stock solution). Label it; it serves all the tests below.
5. Keep a small portion of the original salt untouched — carbonate and ammonium tests belong to it, not to the extract.

3. Neutralisation portions — the working rule

BEFORE TESTING FOR	NEUTRALISE THE EXTRACT PORTION WITH
Sulphate, chloride, bromide, iodide ($AgNO_3$ and $BaCl_2$ tests)	Dilute HNO_3 (boil off CO_2; the solution should end just acidic — test with litmus)
Nitrate (brown-ring test)	Dilute H_2SO_4, then the test itself adds concentrated H_2SO_4
Sulphide, sulphite	Test the original solution or the extract with dilute HCl/H_2SO_4 as the dry test directs — acidification releases the gases

4. The confirmatory wet tests

RADICAL	TEST ON THE (PREPARED) EXTRACT — POSITIVE RESULT
CO_3^{2-}	(Original salt + dilute HCl): effervescence; gas turns lime water milky — and the milkiness disappears on passing excess gas (soluble bicarbonate forms).
SO_4^{2-}	$BaCl_2$ solution → white precipitate ($BaSO_4$) **insoluble** in dilute HCl and in HNO_3 — the insolubility is the confirmation.
Cl^-	$AgNO_3$ → white curdy precipitate (AgCl), soluble in NH_4OH, reappearing on acidifying with HNO_3.
Br^-	$AgNO_3$ → pale yellow precipitate (AgBr), only sparingly soluble in NH_4OH; chlorine water + chloroform/CS_2 → orange-brown organic layer.
I^-	$AgNO_3$ → yellow precipitate (AgI), insoluble in NH_4OH; chlorine water + chloroform → violet organic layer.
NO_3^-	Brown-ring test: extract (or solution) + freshly prepared $FeSO_4$ solution, then concentrated H_2SO_4 poured carefully down the side → a brown ring at the junction of the two liquids.
NO_2^-	Acidify with acetic/dilute H_2SO_4 → brown fumes; $FeSO_4$ + acetic acid gives a brown colour throughout (distinguishing it from the nitrate's ring, which needs concentrated acid).
S^{2-}	Sodium nitroprusside → violet colour; lead acetate paper blackened by the gas from the acidified salt.
SO_3^{2-}	$BaCl_2$ → white precipitate ($BaSO_3$) that **dissolves** in dilute HCl with SO_2 evolved (contrast sulphate); acidified dichromate turns green.
CH_3COO^-	Neutral $FeCl_3$ → deep red colour, turning to a brown precipitate on boiling; warm with concentrated H_2SO_4 + a little ethanol → fruity smell of ethyl acetate.

PO_4^{3-}	Ammonium molybdate in HNO_3, warmed → canary-yellow precipitate.

5. Worked identification (model record)

Dry test indicated a chloride (Experiment 1). Soda extract portion neutralised with dilute HNO_3 and boiled: $AgNO_3$ added → white curdy precipitate; the precipitate dissolved completely in NH_4OH and reappeared on adding HNO_3. **Inference: chloride confirmed.** Note the record's logic — indication, preparation, test, the full behaviour (not merely "white ppt"), conclusion. The precipitate's *behaviour in NH_4OH* is what separates chloride from bromide and iodide, whose precipitates share its colour family but not its solubility.

6. Precautions

- Use distilled water throughout — tap water itself contains chloride and would fake the $AgNO_3$ test.
- Boil the neutralised portion well to expel CO_2 before $BaCl_2$/$AgNO_3$ tests, or carbonate precipitates imitate the result.
- For the brown-ring test the $FeSO_4$ must be freshly prepared and the acid layered without shaking.
- Acidify a fresh portion for each test; never return tested liquid to the stock.

7. Practice (viva preparation)

Answer in your own words; brief model answers follow.

1. Write the double-decomposition that occurs when $ZnSO_4$ is boiled with Na_2CO_3, and state where each product is found after filtration.
2. Why can the soda extract never be used to detect carbonate itself?
3. Three silver precipitates: white fully soluble in NH_4OH; pale yellow sparingly soluble; yellow insoluble. Name the three radicals in order.
4. How do the $BaCl_2$ tests for sulphate and sulphite differ in outcome, and why does the difference confirm?
5. State the brown-ring test procedure and the reason for (a) fresh $FeSO_4$, (b) layering the acid.
6. Why must the extract be boiled after acidifying with HNO_3 before the sulphate test?
7. Give the confirmatory test for acetate with $FeCl_3$, including what happens on boiling.

Answer key

1. $ZnSO_4 + Na_2CO_3 \rightarrow ZnCO_3 \downarrow + Na_2SO_4$. After filtration: zinc carbonate is the residue on the paper; sodium sulphate (the anion, now paired with Na^+) is in the filtrate — the soda extract — ready for

the sulphate test.

2. Because the extract is made with sodium carbonate: carbonate is present from the reagent itself, whatever the salt contained. Carbonate is therefore tested on the original salt with dilute acid (Experiment 1).
3. White, soluble → chloride; pale yellow, sparingly soluble → bromide; yellow, insoluble → iodide. The graded solubility of the silver halides in ammonia is the distinguishing ladder.
4. Sulphate: white $BaSO_4$, insoluble in dilute HCl. Sulphite: white $BaSO_3$, dissolving in dilute HCl with SO_2 given off (dichromate paper greens). Same first appearance, opposite behaviour in acid — behaviour, not appearance, confirms.
5. To the solution add fresh $FeSO_4$ solution; incline the tube and pour concentrated H_2SO_4 slowly down the side so it forms a lower layer; a brown ring appears at the junction. (a) $FeSO_4$ oxidises in air to Fe^{3+}, which cannot play its reducing part. (b) The ring forms only at the undisturbed junction; shaking mixes the liquids and destroys the local conditions (and the observation).
6. To expel dissolved CO_2/carbonate, which with $BaCl_2$ would give white $BaCO_3$ and imitate sulphate; after boiling and acidification, only true $BaSO_4$ can precipitate.
7. Neutral $FeCl_3$ gives a deep red colour (ferric acetate) with acetate; on boiling, the colour discharges into a reddish-brown precipitate of basic ferric acetate.

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